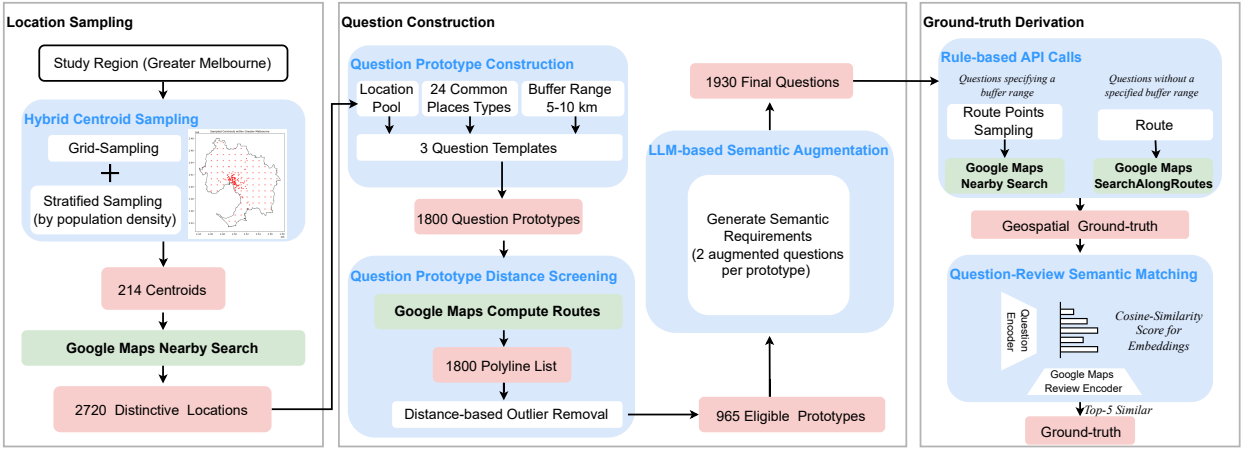
Location Sampling
Study Region (Greater Melbourne)
Hybrid Centroid Sampling
Grid-Sampling
+
Stratified Sampling
(by population density)
214 Centroids
Google Maps Nearby Search
2720 Distinctive Locations
Question Construction
Question Prototype Construction
Location Pool
24 Common Places Types
Buffer Range 5-10 km
3 Question Templates
1800 Question Prototypes
Question Prototype Distance Screening
Google Maps Compute Routes
1800 Polyline List
Distance-based Outlier Removal
965 Eligible Prototypes
LLM-based Semantic Augmentation
Generate Semantic Requirements
(2 augmented questions per prototype)
1930 Final Questions
Ground-truth Derivation
Rule-based API Calls
Questions specifying a buffer range
Route Points Sampling
Google Maps Nearby Search
Questions without a specified buffer range
Route
Google Maps SearchAlongRoutes
Geospatial Ground-truth
Question-Review Semantic Matching
Question Encoder
Cosine-Similarity Score for Embeddings
Google Maps Review Encoder
Top-5 Similar
Ground-truth